

Properties	Axis-angle	Quaternion	Euler angle	Matrix
specific area of usage	Storage of physics variable, i.e. rotational velocity and rotational acceleration, and aithmetic e.g. on collisions or user input.	Calculations of orientation and rotations, specifically sequential combinations and orientation interpolations	Intuitiveness of setting up orientations; e.g. game engine editor windows	Easy integration with other transformations like scaling and translations, thus used in game engine development
mapping to SO(3)	- Repeats every 2π; - For an angle of $n \cdot 2\pi$ any axis represent the identity rotation.	only unit quaternions are rotations; they double cover SO(3)	multiple axis-conventions, repeats every 360° for each entry, suffers from gimbal lock.	1 to 1 mapping of rotation matrices to SO(3), but strict constrictions which matrices represent only rotations
vector rotation	- Rodrigues' rotation formula (requires sin and cos => convert to quaternion in case of multiple rotations)	quaternion sandwich product (32 real multiplication & 18 additions)	not arithmetically possible, requires conversion to matrix or quaternion	matrix-vector multiplication (9 real multiplications & 6 additions) (16 real multiplications & 12 additions for full 4x4 transformation matrices)
airthmetic combinations	- no arithmetical way to calculate sequential rotation combinations - can sum up rotational velocity and acceleration	- sequential rotation combinations via quaternion multiplication (16 real multiplications & 9 additions)	- no sensible arithmetic combinations	- sequential combination of any linear transformation via matrix-matrix multiplication (27 real multiplication & 18 additions for 3x3 rotation matrices; for 4x4 matrices: 64 real multiplications & 48 additions)
interpolation	rotation scaling via lerp of rotation vector; spline interpolation for more control	lerp + normalisation for shortest path; slerp for shortest path & constant speed; squad for control points; spline + normalisation for control points;	lerp interpolation (shortest path if and only if just a single entry differs between interpolation start & end); spline interpolation for more control	Linear interpolation of entries followed by calculating the closest rotation matrix of the interpolation.